Quiz #1: 4 Problems

1. Samyoung Industrial, a manufacturer of table knives and forks, is determining the production volume for each product to maximize profits using its current production facilities. This product is of excellent quality, so they wish to keep it as a manufactured item. Regarding the green process and the next week in which this product is handled, the available workforce for each process is 700 hours for the press process and 1,000 hours for the scroll process. Producing one bag of handles and knives requires 12 minutes (0.2 hours) of press process manpower and 30 minutes (0.5 hours) of scroll process manpower, while producing one case of forks requires 24 minutes (0.4 hours) of press process manpower and 15 minutes (0.25 hours) of scroll process manpower. Samyoung Industrial intends to select the weapons for handles and forks to maximize total profit over the next week. Formulate this problem as a linear optimization model.

Solution to Problem 1

Decision variables and missing information

Let x be the number of bags of knives and y the number of cases of forks produced next week. Let p and q be their respective profits per bag and per case, in won. The question does not provide these unit profits, so the objective must remain symbolic.

Linear optimization model

$$Maximize\ Z = px + qy$$

Subject to

$$0.2x + 0.4y \leq 700$$

$$0.5x + 0.25y \leq 1000$$

$$x \geq 0,\ y \geq 0$$

2. M Asset intends to secure 20 billion won in new funds and invest them in the four stocks in the table. The risk index is a measure of the probability that each stock will deviate from its expected return; a higher value indicates a greater likelihood that the actual return obtained one year from now will deviate from the current expected rate of return. For example, Stock A has the highest expected annual return but also a high-risk index, so the actual return after one year could be much higher than 12%, much lower than 12%, or even result in a loss. On the other hand, Stock C has a low expected return but a very low risk index, so it can be judged that the actual return after one year will be nearly equal to the currently expected 3%. This risk index is a value provided by the Investment Analysis Office. The total risk index of the selected portfolio (Portfolio: distribution of investment funds across stocks) is the sum of the risk indices of the amounts invested in each stock. Management presented investment guidelines stating that, first, the portfolio's expected annual return must be at least 1.8 billion KRW, and second, no more than 50% should be invested in any single stock. Under these guidelines, how should a portfolio be constructed to minimize risk?

Stock	A	B	C	D
Price (10,000 won / Share)	10	5	8	4
Expected Annual Return (%)	12	8	3	10
Risk index per 10,000 won	0.1	0.07	0.01	0.08

Solution to Problem 2

Decision variables and units

Let a, b, c, and d be the amounts invested in Stocks A, B, C, and D, measured in units of 10,000 won. Thus, the budget is 2,000,000 units and the required annual return is 180,000 units.

Linear optimization model

$$Minimize\ R = 0.10a + 0.07b + 0.01c + 0.08d$$

$$a + b + c + d = 2000000$$

$$0.12a + 0.08b + 0.03c + 0.10d \geq 180000$$

$$0 \leq a, b, c, d \leq 1000000$$

3. This problem seeks to find a method for constructing a daily diet that satisfies the daily nutritional requirements while minimizing costs. In this example, we intend to construct a minimum-cost diet using tuna, milk, bread, and spinach to meet the daily requirements for vitamins A, C, D, and iron. The nutritional content and price of each food are shown in the following table. To maintain the taste of the food, at least 0.1 kg of tuna and at least half a loaf of bread must be included. Formulate this problem as a linear optimization model.

Nutritional content	Milk (L)	Tuna (kg)	Bread (ea.)	Spinach (kg)	Daily requirements
Vitamin A (IU)	1600	500	0	70000	5000
Vitamin B (IU)	10	0	0	140	30
Vitamin D (IU)	120	0	0	0	100
Iron (mg)	7	14	13	16	12
Price (Won)	1000	3000	650	600	-

Solution to Problem 3

Decision variables and data clarification

Let m be liters of milk, t kilograms of tuna, b loaves of bread, and s kilograms of spinach consumed per day. The text refers to vitamin C, but the table labels that row vitamin B. The formulation below follows the table; if C was intended, relabel that constraint without changing its coefficients.

Linear optimization model

$$Minimize\ C = 1000m + 3000t + 650b + 600s$$

Minimize daily food cost in won, subject to:

$$1600m + 500t + 70000s \geq 5000$$

Vitamin A requirement (IU).

$$10m + 140s \geq 30$$

Vitamin B requirement, as labeled in the table (IU).

$$120m \geq 100$$

Vitamin D requirement (IU).

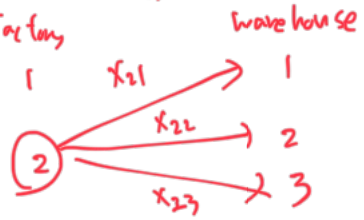
$$7m + 14t + 13b + 16s \geq 12$$

Iron requirement (mg).

$$t \geq 0.1,\ b \geq 0.5,\ m \geq 0,\ s \geq 0$$

4. Company A has two factories located in different regions and intermediate warehouses in three different areas. Each intermediate warehouse requests 500, 2,000, and 900 units of products, respectively, for the following month. Additionally, the two factories can supply up to 2,300 and 1,500 units, respectively, for the following month. The production unit cost at Factory 1 is 150,000 won, while at Factory 2, it is 200,000 won due to high labor costs and outdated facilities. Furthermore, the transportation costs from each factory to the warehouse (unit: 10,000 won/unit) are given in the following table. Company A intends to establish a production and distribution plan to minimize the sum of production costs and transportation costs. Formulate this problem as a linear optimization model.

Factory \ (Storage)	1	2	3
1	3	9	8
2	7	2	4



D.V. $X_{F,W} = X_{\text{Factory, warehouse}}$

Factory F에서 warehouse W로 보내는 운송량

(X_{11}, X_{12}, X_{13}), (X_{21}, X_{22}, X_{23})

$X_{21} + X_{22} + X_{23} \le 1500$

o.f. minize (production cost) + (transportation cost)

① production cost $15(X_{11} + X_{12} + X_{13}) + 20(X_{21} + X_{22} + X_{23})$

② tran— cost. $3X_{11} + 9X_{12} + 8X_{13} + 7X_{21} + 2X_{22} + 4X_{23}$

③ o.f: minimize $(15+3)X_{11} + (15+9)X_{12} + (15+8)X_{13} + (20+7)X_{21} + (20+2)X_{22} + (20+4)X_{23}$

Constraints.

① Demand const.

(warehouse 1) $X_{11} + X_{21} = 500$

(" 2) $X_{12} + X_{22} = 2000$

(" 3) $X_{13} + X_{23} = 900$

② Supply cost.

(Factory 1) $X_{11} + X_{12} + X_{13} \le 2300$

(Factory 2) $X_{21} + X_{22} + X_{23} \le 1500$

③ non-negativity

$X_{11}, X_{12}, X_{13}, X_{21}, X_{22}, X_{23} \ge 0$ (또는 $X_{F,W} \ge 0$ 또는 F, W에 대해서)

Solution to Problem 4

Decision variables and combined unit costs

Let x_{ij} be the units produced at Factory i and shipped to Warehouse j next month, for i = 1, 2 and j = 1, 2, 3. Factory output equals its total shipments.

Measure costs in units of 10,000 won. Production costs are 15 at Factory 1 and 20 at Factory 2. Adding transportation costs gives route costs (18, 24, 23) for Factory 1 and (27, 22, 24) for Factory 2.

Linear optimization model

$$Minimize\ Z = 18x_{11} + 24x_{12} + 23x_{13} + 27x_{21} + 22x_{22} + 24x_{23}$$

Factory capacity constraints:

$$x_{11} + x_{12} + x_{13} \leq 2300$$

$$x_{21} + x_{22} + x_{23} \leq 1500$$

Warehouse demand constraints:

$$x_{11} + x_{21} = 500$$

$$x_{12} + x_{22} = 2000$$

$$x_{13} + x_{23} = 900$$

$$x_{ij} \geq 0 \quad for\ all\ i\ and\ j$$

Demand equalities avoid unnecessary shipments. Using “at least” demand constraints yields the same optimum here because all route costs are positive. Capacity need not be fully used.